

LegalVision: A Knowledge-Driven AI Framework for Legal Understanding and Trust Assessment

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Abstract— Legal documents are often lengthy, complex, and written in highly technical language, making them difficult for citizens and even legal professionals to interpret efficiently. This creates a need for an intelligent legal support system that can improve accessibility, transparency, and trust in document understanding. This study proposes LegalVision, a knowledge-driven AI framework that integrates multi-perspective legal summarization and visualization, explainable legal reasoning, clause-level bias and risk evaluation, and a dynamic legal knowledge graph for property law documents. This research is conducted within the Sri Lankan legal context using a dataset collected from real Sri Lankan legal documents, including property-related deeds and agreements. The framework processes legal texts through clause segmentation, entity and relation extraction, perspective-based summary generation, infographic visualization, risk classification, and graph-supported reasoning, while preserving links to the original clauses for traceability within a single platform. Therefore, this research contributes a unified and explainable legal AI framework that supports both legal professionals and non-expert users in understanding property law documents more accurately and efficiently.

Keywords— *property law, explainable AI, multi-perspective summarization, legal knowledge graph, bias detection, risk classification, trust assessment, visualization*

I. INTRODUCTION

Property-related legal documents play a critical role in ownership transfer, inheritance, leasing, gifting, and other transactions involving immovable property. In Sri Lanka, these documents are commonly written in formal legal language and contain dense clause structures that combine rights, obligations, financial conditions, procedural requirements, and legal consequences within highly technical text. This makes them difficult for non-expert users to interpret and increases the time and effort required by legal professionals to review them accurately. In document-intensive land administration contexts, these challenges reduce accessibility and may affect the quality of legal understanding and decision-making [1].

Recent progress in legal natural language processing has improved several tasks relevant to legal document analysis, including text classification, summarization, and information extraction. Domain-adapted models such as LEGAL-BERT have shown that pretraining on legal corpora can improve

downstream performance compared with general-purpose language models [2]. At the same time, research in AI and law has emphasized that legal support systems must be explainable, since users need to understand not only the output of a system but also the reasoning behind it [3]. This is especially important in legal settings, where trust depends on transparency, traceability, and the ability to justify conclusions using evidence from the original document and relevant legal context [3].

Despite these advances, existing legal AI approaches often address only one part of the legal understanding process. Many studies focus on summarization or classification alone, while offering limited support for interpretable reasoning, clause-level traceability, or the identification of potentially risky and unfair provisions. Prior work has also shown that visual representations can make legal entities, relationships, and document structure easier to understand, particularly for non-expert users [4]. Similarly, automated unfair-clause detection demonstrates the value of machine learning for identifying problematic legal provisions during document review [5]. In the Sri Lankan context, existing digital legal systems are primarily designed for legal professionals with limited accessibility and usability for non-expert citizens, highlighting a gap in user-centered legal support tools.

Another important limitation is the lack of structured legal context. The meaning of a property-related legal document often depends not only on individual clauses, but also on the relationships among parties, assets, obligations, legal acts, and supporting legal sources. Legal knowledge graphs provide a useful way to represent such relationships and support richer retrieval, analysis, and explanation [6], [7]. In property-law document analysis, this kind of structured support can improve contextual interpretation and strengthen traceability across different components of the system.

To address these limitations, this paper presents LegalVision, a knowledge-driven AI framework for legal understanding and trust assessment in Sri Lankan property-law documents. The framework integrates four complementary components: multi-perspective legal summarization and visualization, explainable legal reasoning, clause-level bias and risk evaluation, and a dynamic legal knowledge graph. Rather than treating these tasks independently, the proposed framework combines them to

support a broader process of legal understanding that is more accessible, transparent, and context-aware

The main contribution of this study is the design and evaluation of an integrated framework that helps users understand and trust what a property-related document states, why a particular interpretation is produced, where potential legal risk may exist, and how relevant legal entities and concepts are connected. By combining summarization, reasoning, risk assessment, and graph-based knowledge support within a single architecture, this work contributes toward more trustworthy and user-oriented AI-assisted legal document analysis in the Sri Lankan property-law domain [1], [3], [6].

II. LITERATURE REVIEW

Legal document analysis has become an important research area in natural language processing because legal texts are typically long, domain-specific, structurally complex, and difficult for non-expert users to interpret accurately. These challenges are particularly significant in property-related legal documents, where rights, obligations, ownership details, financial conditions, restrictions, and procedural requirements are often embedded within dense clause structures. As a result, researchers have increasingly explored AI-based methods to improve legal accessibility, efficiency, and interpretability while preserving the legal meaning of the original text [8].

One major area of prior research is legal text summarization, which aims to reduce the burden of analyzing lengthy legal documents by presenting essential information in a concise and accessible form. Early approaches primarily relied on extractive techniques, using statistical or rule-based methods to select key sentences from the original text. While these methods preserved the original legal wording, they often produced fragmented summaries and failed to capture the broader semantic meaning and contextual relationships within the document. More recent transformer-based approaches have improved performance by enabling abstractive summarization and better contextual understanding of long texts [8], [9]. However, most existing systems focus on generating a single, general-purpose summary. In practice, legal users require information from multiple perspectives, such as ownership, duties, restrictions, liabilities, and financial implications. Additionally, visualization remains underexplored, as current systems provide limited or oversimplified graphical representations that fail to convey relationships, dependencies, and proportions within legal content. These limitations highlight the need for multi-perspective summarization approaches supported by accurate and context-aware visualizations.

A second important area is explainable legal reasoning. Legal AI systems are increasingly expected not only to produce outputs, but also to justify them in a way that users can understand and trust. This requirement is especially important in legal environments, where conclusions must be supported by interpretable reasoning and traceable evidence. Prior studies in AI and law emphasize that transparency is essential for responsible legal decision support, particularly when systems assist with interpretation, classification, or recommendation tasks [3]. Recent work on large language models in the legal domain also demonstrates strong potential for automated legal assistance, including reasoning and question answering, but highlights continuing concerns

related to hallucination, limited traceability, and lack of domain-grounded explanation [10]. These limitations suggest that accuracy alone is insufficient in legal contexts. A useful legal reasoning system must also show why a conclusion was reached, what textual or legal evidence supports it, and how the reasoning can be validated by the user.

Another important line of research focuses on bias, unfairness, and risk analysis in legal text. Legal documents may contain provisions that are one-sided, ambiguous, restrictive, or otherwise problematic, even when they appear formally valid. This has motivated the development of machine learning approaches for identifying unfair contract terms, imbalanced clauses, and legally sensitive language patterns. Studies such as CLAUDETTE demonstrate that automated classification can assist in detecting potentially unfair clauses and improving document review processes [5]. Other work in legal text analytics has shown that NLP-based clause-level analysis can help identify liabilities, warranties, obligations, and problematic contractual conditions [11], [12]. Despite these advances, many prior systems focus on narrow forms of classification and do not extend their outputs into broader trust assessment. In the context of property-law documents, this is a significant limitation. Users may need support not only in identifying the clause type, but also in understanding whether a clause introduces legal uncertainty, asymmetric obligations, or practical risk. This indicates the need for a more user-oriented evaluation layer that treats clause analysis as part of trust assessment rather than as an isolated prediction task.

A further relevant research direction is the use of legal knowledge graphs. Knowledge graphs provide structured representations of entities, legal concepts, and relationships, making them useful for organizing complex legal information in a form that supports semantic search, contextual retrieval, and interpretable analysis. Prior work has shown that graph-based legal representations can improve the ability to connect statutes, concepts, obligations, actors, and other legal elements across documents [6], [7]. Compared with purely text-based systems, legal knowledge graphs offer better support for exploring how legal information is related, which is particularly valuable for users who need contextual understanding rather than isolated answers. However, many existing legal knowledge graph approaches are either general-purpose or focused on case law, regulation, or statutory reasoning. Their direct application to property-law document interpretation remains limited. In addition, graph-based methods are often developed as standalone resources rather than being integrated into end-user legal assistance systems. This reduces their practical value in settings where users need summaries, reasoning support, risk awareness, and contextual structure together.

Across these four research areas, the literature demonstrates substantial progress in supporting legal information processing. Summarization improves accessibility, explainable reasoning improves interpretability, unfair-clause detection improves document review, and knowledge graphs improve contextual representation. However, most existing studies address these capabilities independently. This creates a gap between the outputs of legal AI models and the broader needs of users who must understand, evaluate, and trust legal documents in realistic settings. A user reviewing a property-related legal document may require a concise summary, an explanation of legal

implications, identification of potentially risky clauses, and a structured view of how legal entities and concepts are connected. Existing solutions rarely combine these requirements within a single framework.

Another gap in the literature is the limited attention given to Sri Lankan property-law documents. Much of the existing research in legal NLP has focused on English-language case law, contracts, statutes, or legal benchmarks drawn from Western jurisdictions [2], [8], [11]. In contrast, property-related legal documents in Sri Lanka involve context-specific legal terminology, formal clause patterns, and document structures that are less represented in current legal AI research. This lack of domain-specific focus reduces the direct applicability of many existing models and highlights the need for frameworks tailored to local legal documentation practices.

Therefore, the literature suggests two important conclusions. First, individual advances in summarization, explainability, clause analysis, and structured legal knowledge are valuable but insufficient when applied in isolation. Second, there is a need for integrated and domain-aware legal AI systems that support not only automated analysis, but also user understanding and trust. In response to these gaps, this study proposes LegalVision, a knowledge-driven AI framework that combines multi-perspective summarization, explainable legal reasoning, clause-level bias and risk evaluation, and dynamic knowledge graph construction for Sri Lankan property-law documents. By integrating these components into a unified system, the proposed framework aims to move beyond single-task legal AI and provide a more complete foundation for legal understanding and trust assessment.

III. METHODOLOGY

A. Dataset Creation

The dataset for this research was created using a primary-data-driven approach centered on Sri Lankan property-law documents. The initial raw corpus was collected with the support of Sri Lankan lawyers and notaries, ensuring that the documents reflected authentic legal drafting styles, terminology, and clause structures found in practice. To preserve privacy and ethical compliance, sensitive personal details in real deeds were anonymized before inclusion in the research dataset. In addition to real anonymized deeds, synthetic deeds were also prepared to represent rare legal structures and edge cases that were not sufficiently available in the collected corpus.

The study focuses on a set of common property-related deed categories, including sale deeds, deeds of gift, last wills, leases, mortgage-related instruments, and transfer agreements. These raw documents were then reviewed and annotated to create task-specific datasets for the four project components. First, a clause-level dataset was created by segmenting deeds into individual clauses and assigning each clause to a legal perspective such as ownership and parties, financial impact, conditions and procedures, or rights and risks. Second, a multi-perspective summary dataset was prepared by grouping relevant clause content and writing structured summaries under each perspective. Third, a legal jargon dictionary was developed to map complex legal terms into simplified meanings and synonyms for plain-language support. Fourth, a risk and bias dataset was prepared by labeling clauses with risk type, bias category, and severity indicators. Supporting

resources such as survey-plan references, administrative divisions, and Sri Lankan property-law statutes were also collected to improve contextual interpretation and downstream reasoning.

TABLE summarizes the dataset components used in the LegalVision framework. The current analysis identifies 1,707 clause records, 320 multi-perspective summaries, and 365 legal terms. As shown in TABLE, these datasets are interconnected resources that support summarization, legal reasoning, bias/risk analysis, and knowledge-graph construction.

TABLE I. DATASET COMPONENTS USED IN THE LEGALVISION FRAMEWORK

Dataset Component	Description	Source	Purpose
Clause Dataset	Clause-level deed segments	Real deeds	Perspective classification, risk analysis
Summary Dataset	Multi-perspective summaries	Annotated deeds	Perspective-based summarization
Legal Jargon Dictionary	Legal terms with simplified meanings	Domain experts + statutes	Plain-language explanation
Deed Corpus	Property-law deeds	Lawyers/notaries	Source documents
Legal Source Collection	Statutes, regulations, supporting resources	Legal references	Reasoning and graph support

B. Dataset Preprocessing

After collection, the raw legal documents were converted into machine-readable text using a document processing pipeline that supports formats such as PDF, DOCX, and TXT. The extracted text was then cleaned to remove formatting noise, broken lines, duplicated symbols, and other inconsistencies caused by scanning or digital conversion. Because legal deeds often contain long paragraphs and embedded metadata, special attention was given to clause segmentation, which is a key preprocessing step in the system. Each deed was split into clause-level units so that later stages such as summarization, reasoning, and risk classification could operate on meaningful legal segments rather than on the full document at once.

The next stage involved normalization and annotation. Deed type labels were standardized, clause IDs and deed IDs were assigned, and perspective labels were attached to clauses based on annotation. In parallel, legal jargon terms found in the raw deed text were matched against the custom dictionary so that the system could retain the original legal wording while also generating simplified explanations for non-expert users. This step is especially important because system is not only intended to analyze documents, but also to improve accessibility through plain-language support.

Further preprocessing included named entity recognition and structured field extraction. Important entities such as party names, NIC numbers, deed types, dates, consideration values, plan numbers, lot numbers, boundary references, extents, and registry details were identified from the deed corpus. These extracted elements were used both for structured summarization and for later knowledge-graph construction and also for the infographic generation. For the bias/risk component, annotated clauses were transformed into

supervised training samples with labels such as bias type, risk level, and bias score. For the reasoning component, supporting legal sources such as statutes and decision trees were prepared in structured form so that the system could later generate traceable legal reasoning paths.

C. Model Architecture

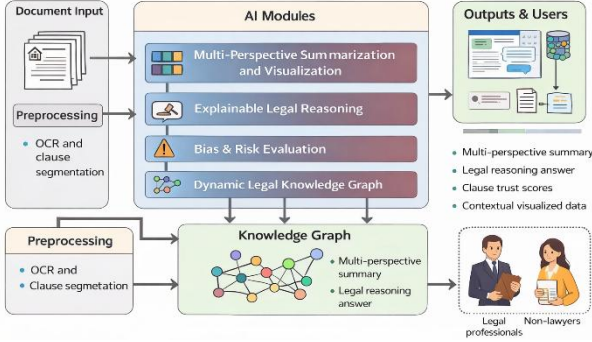


FIGURE 1. OVERALL ARCHITECTURE

The proposed architecture follows a modular, integrated design in which all four research components operate on top of a shared document-processing pipeline. At the input stage, the system accepts uploaded legal documents and performs text extraction, clause segmentation, normalization, and entity-aware preprocessing. These structured outputs are then passed to the four main components: multi-perspective summarization and visualization, explainable legal reasoning, bias and risk evaluation, and the dynamic legal knowledge graph. This shared architecture is important because the system is not intended to produce isolated outputs, but rather a connected understanding of the document that is transparent, evidence-based, and useful for both legal professionals and non-expert users.

The multi-perspective summarization and visualization module is designed to convert dense property-law content into user-relevant summaries supported by visual aids. Clauses are first encoded using LegalBERT-based contextual representations and grouped by legal perspective. A sequence-to-sequence summarization model then generates summaries for each perspective from the grouped clause content with mapping jargon-normalized words. These outputs are further connected to a visualization layer that can display timelines, clause groupings, infographics and other explanatory visual formats to make the summarized content easier to understand. Because the summaries are generated from clause-grouped inputs, the module supports traceability between the final summary and the original legal text.

The explainable legal reasoning module is built around prompt chaining, argumentation structures, and large language model support. This module generates step-by-step, auditable reasoning chains for Sri Lankan property law and links each insight to supporting statutes, precedents, and document clauses. Rather than behaving like a black-box answer generator, the model architecture is designed to show how a legal conclusion is formed. Logic trees and visual reasoning graphs are used to improve interpretability, while structured legal sources provide grounding for legally accountable outputs. This makes the reasoning component central to the transparency goal of the whole system.

The bias and risk evaluation module uses a fine-tuned transformer-based multi-label classifier at clause level. The proposal specifies that the system accepts raw legal text,

segments it into clause units, and predicts both property-law-specific bias categories and risk severity levels. It also identifies bias direction, such as whether a clause favors one party over another, and combines these outputs into a calibrated document-level trust score between 0.0 and 1.0. The module further includes an explainability layer in which model-derived importance signals are merged with concise rule-based rationales, allowing flagged clauses to be shown with color-coded highlights and tooltips. From an architectural point of view, this module combines classification, calibration, explanation, and user-facing legal triage in a single workflow.

The Dynamic Legal Knowledge Graph module serves as the structural backbone of the system, enabling efficient organization and retrieval of legal data. It integrates NLP techniques, particularly a spaCy-based NER model, to extract entities such as parties, property details, financial values, and locations from legal documents. These entities are linked through relationship extraction and stored in Neo4j as nodes and relationships (e.g., SOLD, BOUGHT, LOCATED_IN). The methodology includes data preprocessing, ontology design, entity and relationship extraction, and graph construction using Cypher. A layered architecture with React, FastAPI, and Neo4j supports visualization, processing, and storage. This graph enables semantic search, contextual linking, and supports advanced reasoning within the system.

Overall, the methodology is built around real legal data collection, annotation, domain-specific preprocessing, and a shared multi-module architecture. The summarization module improves accessibility, the reasoning module improves transparency, the bias and risk evaluator improves trust assessment, and the legal knowledge graph improves contextual and semantic understanding. By integrating these components within a common pipeline, the methodology supports a more complete and trustworthy approach to AI-assisted analysis of Sri Lankan property-law documents.

D. Evaluation Strategy and Performance Metrics

The LegalVision framework was evaluated at the component level because it consists of four distinct modules. Instead of using a single overall measure, each module was assessed using task-specific metrics suited to its objective.

For the multi-perspective summarization and visualization module, evaluation was conducted in two stages. The clause classification task was assessed using accuracy and macro F1-score, while the generated summaries were evaluated using ROUGE metrics to measure overlap with reference summaries.

For the bias and risk evaluation module, the clause-level classification task was evaluated using accuracy and F1-score. These metrics were selected to measure the model's effectiveness in identifying legally sensitive or potentially risky clauses.

For the explainable legal reasoning module, evaluation focused on the quality of generated legal explanations. BLEU and ROUGE were used to assess textual similarity, while BERTScore was used to measure semantic similarity between generated and reference outputs.

For the dynamic legal knowledge graph module, evaluation focused on the named entity recognition stage used for graph construction. Precision, recall, and F1-score were

used to measure the accuracy of legal entity extraction before storage in the graph database.

Overall, this evaluation strategy ensured that each module was assessed using the most appropriate performance measures for its task, providing a clear basis for analyzing the effectiveness of the LegalVision framework.

IV. RESULTS AND DISCUSSION

The proposed LegalVision framework was evaluated at the component level because it consists of four modules: multi-perspective summarization and visualization, explainable legal reasoning, bias and risk evaluation, and a dynamic legal knowledge graph. This evaluation approach matches the overall system design, where all four modules work together to improve accessibility, transparency, contextual understanding, and trust in property-law document analysis.

A. Results of the Multi-Perspective Summarization and Visualization Module

The summarization and visualization module was evaluated in two stages: clause-level perspective classification and summary generation. In the classification stage, LegalBERT achieved an accuracy of 94.86% and a macro F1-score of 94.49%, indicating strong performance in assigning clauses to the four legal perspectives. This result is important because accurate clause grouping directly affects the quality of downstream summary generation.

TABLE II. PERFORMANCE OF CLAUSE-LEVEL PERSPECTIVE CLASSIFICATION USING LEGALBERT

Model	Task	Accuracy	Macro F1-score
LegalBERT	Perspective classification	94.86%	94.49%

The summarization results of LegalLED indicate that the model was able to generate useful perspective-based summaries while preserving key legal content. Compared with generic summarization tasks.

TABLE III. EVALUATION RESULTS OF MULTI-PERSPECTIVE SUMMARIZATION USING LEGALLED

Model	ROUGE-1	ROUGE-2	ROUGE-L	ROUGE-Lsum
LegalLED	0.4204	0.2367	0.3303	0.3313

Further comparison with T5 showed that LegalLED was better suited to this task due to its longer input capacity and stronger support for long-document processing.

Qualitative outputs showed that the module could generate structured summaries for perspectives such as Ownership and Parties Involved, Financial and Asset Impact, Conditions and Procedures, and Rights, Duties, and Risks. These results confirm that the module improves accessibility by transforming lengthy deed content into more organized and interpretable outputs.

B. Results of the Bias and Risk Evaluation Module

The bias and risk evaluation module achieved 88% accuracy and 88% F1-score, showing strong performance in detecting clauses that may be one-sided, ambiguous, or legally

risky. A comparison with DistilBERT, which achieved 83% accuracy, confirmed that the selected model performed better.

TABLE IV. COMPARISON OF BIAS AND RISK CLASSIFICATION PERFORMANCE

Model	Accuracy	F1-score
DistilBERT	83%	—
Proposed Risk Model	88%	88%

The interface results further showed the usefulness of this component. Clauses were flagged with labels such as unilateral termination, financial imbalance, and risk levels. This demonstrates that the system does more than simple classification, as it provides interpretable outputs that highlight the type and severity of legal risk.

This module adds an important trust-assessment layer to the framework. While summarization explains what a document says, the risk model highlights what may be problematic. Users can use these outputs as decision-support indicators rather than final legal judgments.

C. Results of the Explainable Legal Reasoning Module

The explainable legal reasoning module achieved strong performance across multiple metrics (Table V), producing accurate and semantically consistent legal explanations.

TABLE V. EVALUATION RESULTS OF THE EXPLAINABLE LEGAL REASONING MODULE

Metric Type	Metric	Score
BLEU	BLEU-1	73.4
BLEU	BLEU-2	56.3
BLEU	BLEU-3	46.7
BLEU	BLEU-4	38.8
BLEU	Overall BLEU	52.6
ROUGE	ROUGE-1	0.707
ROUGE	ROUGE-2	0.472
ROUGE	ROUGE-L	0.590
BERTScore	Precision	0.931
BERTScore	Recall	0.932
BERTScore	F1	0.931

The qualitative outputs showed that the system could provide structured answers to practical legal questions, including estimated costs, required documents, reasons, consequences, and cost breakdowns. This demonstrates that the module can generate transparent and user-friendly legal guidance rather than brief black-box responses.

Among the four components, this module appears to be one of the strongest. Its good performance is likely supported by structured prompts and organized answer generation. Still, its output depends on the quality of legal sources and upstream extraction, so it should be used as a support tool rather than a substitute for professional legal advice.

D. Results of the knowledge graph Module

The knowledge graph component produced the strongest extraction results in the system. The custom NER model achieved 99.19% precision, 99.87% recall, and 99.53% F1-score, showing that it was highly effective in identifying important deed-related entities required for graph construction.

TABLE VI. PERFORMANCE OF THE KNOWLEDGE GRAPH ENTITY EXTRACTION MODULE

Model	Precision	Recall	F1-score
Custom SpaCy NER	99.19%	99.87%	99.53%

The Neo4j-based implementation further showed that the extracted entities were successfully stored and connected in a relational graph structure. The graph interface supported interactive visualization, instance-level storage, and query-based exploration of legal entities and their relationships.

This component acts as the structural backbone of the framework by supporting context-aware analysis and semantic linking. Very high recall is especially important because missing entities would weaken graph completeness and downstream reasoning. However, graph usefulness still depends on the richness of relationships, query quality, and continuous updates.

E. Overall Discussion

When all four components are considered together, several key findings emerge. First, the strongest performance was observed in structured tasks, especially clause classification and graph-based entity extraction. Second, the reasoning module also performed strongly, particularly in semantic accuracy and practical answer quality. Third, the summarization module, although effective, remained the most challenging because it must balance compression, readability, and legal faithfulness.

A major contribution of this work is the integration of all four components into a single framework for Sri Lankan property-related legal documents. Unlike many existing studies that focus on only one task, LegalVision combines summarization, reasoning, risk evaluation, and graph-based knowledge support within one system.

Some limitations should also be noted. The datasets are still relatively small compared with large international benchmarks, especially for summarization and bias detection. Some annotations may contain subjective variation, and the system currently focuses only on selected property-related deed categories. In addition, errors in preprocessing may affect downstream modules.

Overall, the results show that LegalVision is technically feasible and performs strongly across its main components. The framework achieved very high results in classification and knowledge graph extraction, strong semantic performance in legal reasoning, and promising results in multi-perspective summarization. These findings suggest that the system can provide more accessible, transparent, and trustworthy AI-assisted analysis of Sri Lankan property-related legal documents.

V. CONCLUSION AND FUTURE WORK

This study presented LegalVision, an AI framework for analyzing Sri Lankan property-related legal documents through multi-perspective summarization and visualization, explainable legal reasoning, bias and risk evaluation, and a

dynamic legal knowledge graph. The results show that the framework can improve legal document understanding by making content more accessible, transparent, and structured for both legal professionals and non-expert users.

The main advantage of this research is the integration of multiple legal AI functions within one system. Instead of only summarizing documents, the framework also explains legal meaning, identifies risky clauses, and connects legal entities through structured graph-based knowledge.

Future work can extend this research in several directions. Beyond property law, an important future direction is to adapt the framework to other legal domains, such as contract law, family law, criminal law, land dispute documentation, labor law, and administrative law. Extending LegalVision to these areas would allow the framework to support a wider range of legal document types, legal reasoning patterns, and domain-specific risks. In the long term, this research has the potential to improve legal accessibility, strengthen trust in AI-assisted legal systems, and contribute to more practical and socially useful legal technology solutions in Sri Lanka.

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